RodentNet

Privacy-Preserving Federated Learning for Collaborative Immunology

Group 8
Sager Prasad (B23CSB56)
Harishankar S (B23CSB34)
Angelina Rose M (B23CSB19)
Anaswara S P (B23CSB17)

Project Guide: Dr. Thushara A
External Guide: Mr. Gokul Alex

October 2026

Contents

1. Introduction
2. Related works
3. Problem statement and objectives
4. Relevant SDGs
5. Methodology
6. Implementation status
7. Conclusion
8. References

Introduction

- Federated learning (FL) trains a model across institutions while keeping raw data local.
- RodentNet targets collaborative healthcare and immunology research.
- Learn from distributed T-cell receptor (TCR) and epitope sequences without pooling records.
- CDR3 sequences act as genomic barcodes, so HIPAA/GDPR block centralization.

Challenges

- Non-IID client data (skewed epitope distributions).
- Leakage from individual updates (gradient inversion, membership inference).
- Update tampering and impersonation by an active network adversary.
- "Harvest now, decrypt later": quantum threats to DH, RSA and Ed25519.
- Client dropout: stranded masks, and forward-secrecy leaks when recovering them.
- Security overhead and coordination.

Related works I: secure federated aggregation

Paper	Year	Contribution	Limitation / gap
PQSF: post-quantum secure privacy-preserving FL [6]	2024 Zhang et al.	Lattice-based multi-stage secret sharing and double masking.	Cryptographic aggregation only; no personalization or TCR task.
Multi-key aggregation for FL [7]	2024 Li et al.	Homomorphic encryption with proxy re-encryption under different keys; dropout support.	Two non-colluding semi-honest servers; ciphertext operations; no PQ authentication.
FL via MPC and compressed sensing [8]	2024 Chen et al.	Secret sharing with functional encryption; compressed sensing cuts cost.	Gradient protection and efficiency; no PQC authentication or immunology.
Beskar [15]	2025 Zhang et al.	Full-stack PQ secure aggregation; setup amortized over rounds.	Relies on designated assisting nodes to unmask dropped clients.

Related works II: personalization and TCR prediction

Paper	Year	Contribution	Limitation / gap
CosPer: adaptive personalization [9]	2024 Ren et al.	Cosine similarity of global and local gradients adapts aggregation; fairness and robustness.	No cryptographic update confidentiality; no MPC or PQC.
MixTCRpred [10]	2024 Croce et al.	Epitope-specific predictor from 17,715 paired $\alpha\beta$ TCRs.	Centralized training; no federation or secure aggregation.
DKD-pFed [11]	2025 Su et al.	Decoupled distillation and feature decorrelation for heterogeneous clients.	Image benchmarks; no integrated MPC/PQC or sequence task.

Related works III: post-quantum and healthcare FL

Paper	Year	Contribution	Limitation / gap
PQS-BFL [12]	2026 Commey, Crosby	ML-DSA-65 authentication with blockchain verification.	Needs blockchain; no threshold/pairwise-MPC combination for TCR data.
Secure aggregation on clinical EEG [13]	2026 Rajabi, Toorani	Masking, threshold sharing, dropout recovery in Flower.	EEG data; no ML-DSA authentication or TCR models.
PQ FL for cyber defense [14]	2025/26 Nayak et al.	Kyber key establishment and Dilithium authentication.	Threat intelligence, not immunology.
PQ signatures over FL [16]	2024 Zhang et al.	Signature overhead benchmark; verification fast, payload grows with clients.	Benchmark only; motivates session-key amortization.

Problem statement and objectives

Problem statement

Develop a secure and generalizable cross-institutional machine-learning framework for TCR–epitope prediction that enables collaborative learning under privacy constraints, heterogeneous experimental protocols and non-IID datasets, with quantum-resilient protection of participation and model updates.

Objectives

1. Enable cross-institutional learning while keeping raw TCR and immunological data local.
2. Handle non-IID hospitals with Dirichlet-skewed VDJdb; compare FedAvg, FedPer, fine-tuning, local-only and pooled training.
3. Learn TCR–epitope specificity with a 1D-CNN and FastKAN head under a strict non-overlapping clonotype split.
4. Protect participation and updates with ML-DSA-65, ML-KEM-768 and self-masked pairwise aggregation over $\mathbb{F}_p$.
5. Recover from dropouts without weakening forward secrecy, using ephemeral Shamir-shared round seeds.
6. Evaluate utility, security and overhead over 10 seeds against heuristic and federated baselines.

Relevant Sustainable Development Goals

SDG 3: Good Health and Well-Being

Target 3.b: support vaccine and medicine research.

Contribution: collaborative immunology research with TCR–epitope models.

SDG 9: Industry, Innovation and Infrastructure

Target 9.5: strengthen scientific research and technological capabilities.

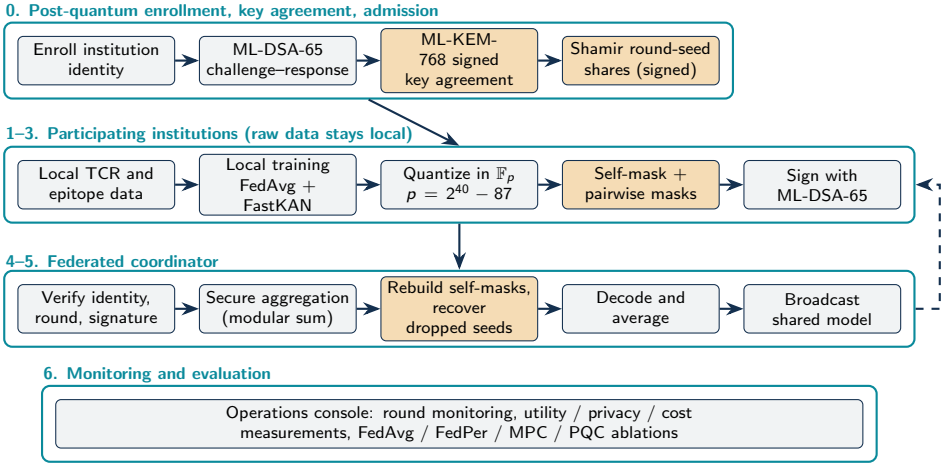
Contribution: privacy-oriented, quantum-resilient, authenticated research infrastructure.

SDG 17: Partnerships for the Goals

Target 17.6: advance science and technology cooperation.

Contribution: inter-institutional learning without centralizing raw data.

Methodology



Amber: new since the previous review. Dashed arrow: synchronized model returns to institutions.

Methodology: post-quantum enrollment and key agreement

Shamir threshold custody. A signing secret s is embedded in a random polynomial:

$$f(z) = s + \sum_{\ell=1}^{t-1} a_\ell z^\ell \pmod{q}, \quad \text{share}_i = (i, f(i)), \qquad s = \sum_{i \in S} f(i) \prod_{j \in S, j \neq i} \frac{-j}{i-j}.$$

Challenge–response with ML-DSA-65.
$m_k = id_k \,\|\, nonce \,\|\, t,\ \sigma_k = \mathrm{Sign}_{sk_k}(m_k),\ \mathrm{Verify}_{pk_k}(m_k, \sigma_k) = 1.$

Pairwise key agreement. Clients $i < j$ agree on a 32-byte root secret s_{ij} with ML-KEM-768; the signature covers (ciphertext $\|$ pk $\|$ session), so a relay cannot substitute a ciphertext.

- Nonce prevents replay; identity and round binding prevent cross-client or cross-round reuse.
- Sizes: ML-KEM-768 pk 1,184 B, ciphertext 1,088 B; ML-DSA-65 pk 1,952 B, signature 3,293 B (Round-3; 3,309 B in FIPS 204).
- Key exchange is paid once in Round 0 and amortized over all rounds.

Methodology: local datasets and participating institutions

Client k retains a local dataset of paired sequences:

$$D_k = \{(x_{ki}^{TCR}, x_{ki}^{E}, y_{ki})\}_{i=1}^{n_k}, \qquad D_k \sim P_k,\ P_i \neq P_j \text{ in general.}$$

Amino-acid sequences map to fixed-length token vectors $\psi : \mathcal{A}^* \to \{\text{PAD}, \text{UNK}, 1, \ldots, 20\}^L$. The collaborative objective weights institutions by sample count:

$$F(\Theta) = \sum_{k=1}^{K} \alpha_k F_k(\Theta), \qquad \alpha_k = \frac{n_k}{\sum_{j=1}^{K} n_j}.$$

- Non-IID split: per-hospital label proportions drawn from $\text{Dirichlet}(\alpha = 0.5)$ across $K = 3$ hospitals.
- Strict split: 3,091 training and 773 test clonotypes (80/20) with zero CDR3 overlap.
- Raw D_k never leaves client k; only masked, signed updates are exchanged.

Methodology: federated training (FedAvg and FedPer)

Each client runs local gradient steps in round t:

$$\theta^t_{k,e+1} = \theta^t_{k,e} - \eta \nabla L_k(\theta^t_{k,e}), \qquad \Delta\theta^t_k = \theta^t_{k,E} - \theta^t.$$

FedAvg (primary). The whole model is shared and averaged: $\theta^{t+1} = \theta^t + \sum_k \alpha_k \Delta\theta^t_k$.
FedPer (compared). Shared θ, client-specific head ϕ_k:

$$\min_{\theta,\{\phi_k\}} \sum_{k=1}^{K} \alpha_k L_k(\theta, \phi_k), \qquad \phi^{t+1}_k = \phi^t_{k,E} \text{ stays local.}$$

- Under Dirichlet skew, hospitals hold few rare-epitope examples, so local heads overfit.
- Measured: FedAvg 39.35% vs FedPer 33.07% balanced accuracy (global test).

Methodology: the TCR–epitope learning method

Top-10 classifier (187,466 parameters): token embedding (dim 64) $\rightarrow$ 1D-CNN (2 layers, 128 channels, kernel 3) $\rightarrow$ FastKAN $\rightarrow$ 10-way softmax. The Siamese stress model (251,878 parameters) uses dual encoders $h_i^T = f_{\theta_T}(x_i^{TCR})$, $h_j^E = f_{\theta_E}(x_j^E)$. A FastKAN layer replaces linear maps by radial basis functions:

$$\phi(x) = \sum_{k=1}^{G} c_k \exp\Big(- \frac{(x - \mu_k)^2}{2\sigma^2} \Big).$$

The Siamese variant scores pairs with a FastKAN head and a temperature τ, trained with symmetric InfoNCE:

$$z_{ij} = \mathrm{FastKAN}([h_i^T \parallel h_j^E])/\tau, \qquad L_{NCE} = \tfrac{1}{2}\big[\mathrm{CE}(Z, I_B) + \mathrm{CE}(Z^\top, I_B)\big].$$

- Diagonal entries are matched pairs; off-diagonal entries act as in-batch negatives.
- The Top-10 classifier carries the headline federated results; the Siamese model is the upper-bound stress model.

Methodology: finite-field quantization and masking

Field and scale: $p = 2^{40} - 87 = 1{,}099{,}511{,}627{,}689$, $S = 10^6$. Weight and quantize the update:

$$q_k^t = \mathrm{round}\big(S\, n_k \Delta\theta_k^t\big) \bmod p, \qquad \mathrm{decode}(z) = z/S \text{ if } z < p/2,\ (z-p)/S \text{ otherwise.}$$

Pairwise masks from the ML-KEM root, plus a private self-mask $b_i^{(r)}$:

$$M_{ij}^{(r)} = \mathrm{SHAKE256}(\text{“pairwise-seed”}, i, j, r, s_{ij}) \bmod p,$$

$$y_k^t = q_k^t + M_k^{(self)} + \sum_{j>k} M_{kj}^t - \sum_{j<k} M_{jk}^t \pmod p.$$

- No overflow: $|w| \le 10$, $N = 100 \Rightarrow \sum |enc| \le 10^9 \ll p/2 \approx 5.49 \times 10^{11}$.
- The self-mask defeats false-dropout attacks: a falsely “dropped” client stays blinded.
- Bound to sender and round: $m_k^t = id_k \| t \| y_k^t$, $\sigma_k^t = \mathrm{Sign}_{sk_k}(m_k^t)$.

Methodology: authenticated secure aggregation

Accept only valid, round-bound uploads: $A^t = \{k : \mathrm{Verify}_{pk_k}(m_k^t, \sigma_k^t) = 1\}$.

Pairwise masks between survivors cancel; survivors release shares of $b_i^{(r)}$ so the server removes the self-masks:

$$\sum_{k \in A^t} y_k^t - \sum_{k \in A^t} M_k^{(self)} \equiv \sum_{k \in A^t} q_k^t \pmod{p}.$$

Decode the weighted sum and update the shared model:

$$\theta^{t+1} = \theta^t + \frac{\mathrm{Decode}\left(\sum_{k \in A^t} q_k^t\right)}{\sum_{k \in A^t} n_k}.$$

- A missing or rejected client leaves unmatched masks; they are recovered by forward-secret dropout recovery (next slide).
- The server recovers the aggregate $\sum W_i$, never an individual update.

Methodology: forward-secret dropout recovery

1. **Ratchet:** $s_{ij}^{(r+1)} = \mathrm{SHAKE256}(\text{“root-ratchet”}, s_{ij}^{(r)})$ (one-way).
2. **Derive:** round seed $K_{ij}^{(r)}$ by domain-separated SHAKE-256.
3. **Share:** split $K_{ij}^{(r)}$ with a (t, N) Shamir polynomial, $t = \lfloor N/2 \rfloor + 1$; each share is signed with ML-DSA-65.
4. **Recover:** if client k drops, survivors send their shares of $K_{kj}^{(r)}$; the server interpolates

$$K_{kj}^{(r)} = \sum_{m=1}^{t} f(m) \prod_{\ell \neq m} \frac{-\ell}{m - \ell} \pmod{257}.$$

5. **Unmask:** regenerate client k's stranded masks for round r and subtract them.

- Forward secrecy: learning $K_{kj}^{(r)}$ reveals nothing about past seeds or $s_{kj}^{(r+1)}$ (preimage resistance).
- Each pairwise seed is shared independently (GF(257), 37 B per share): Kyber encapsulation is directional, so a dropped client's key cannot recover K_{AB}.

Methodology: broadcast and repeat federated rounds

The coordinator broadcasts the synchronized shared parameters:

$$\theta_{k,0}^{t+1} \leftarrow \theta^{t+1} \quad \text{(FedPer: } \phi_k^{t+1} \text{ stays local).}$$

$$\theta^t \xrightarrow{\text{local SGD}} \theta_{k,E}^t \xrightarrow{\text{mask, sign, upload}} \xrightarrow{\text{secure aggregation}} \theta^{t+1}.$$

After $R = 10$ rounds, the deployed client model is $M_k^{(R)}(x) = M(x; \theta^R)$.

- Every institution receives collaborative knowledge through θ^R.
- Per round, clients re-derive fresh masks from the ratcheted root secrets; no new public-key exchange is needed.
- The cycle repeats without exchanging raw TCR or epitope records.

Methodology: monitoring, evaluation and ablation

Class-balanced metrics over $C = 10$ epitopes (and retrieval metrics for the Siamese model):

$$\text{BalAcc} = \frac{1}{C}\sum_{c=1}^{C}\text{recall}_c, \qquad \text{MRR} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\text{rank}_i}.$$

Round cost separates learning from protection overhead:

$$T_{round} = \max_k T_{train,k} + T_{mask} + T_{sign} + T_{verify} + T_{aggregate}, \qquad C_{round} = \sum_{k\in A^t}(|y_k^t| + |\sigma_k^t| + |\text{shares}_k|).$$

Ablation relative to the full system: $\Delta M_{-c} = M_{full} - M_{without\ c}$.

- Baselines: random, majority, label-shuffle, CDR3 1-NN, 3-mer LogReg, pooled, local-only, FedAvg + fine-tuning, FedPer.
- 10 seeds (42–51); gains reported as paired differences with 95% CIs and paired t-tests.
- The React/FastAPI console shows metrics and handshake status by round.

Methodology: threat model and non-goals

Defended

- Semi-honest aggregator inspecting updates and transcripts.
- Up to $t-1$ colluding clients, $t = \lfloor N/2 \rfloor + 1$.
- Active MITM: replay, drop, substitute keys and ciphertexts.
- Quantum "harvest now, decrypt later" adversary.
- Abrupt client dropout at any time.

Explicit non-goals

- Malicious data poisoning: needs range proofs or norm clipping.
- Leakage from the revealed aggregate $\sum W_i$: needs differential privacy.
- Active view-partitioning by a malicious server: needs signed survivor-consistency rounds with $t > 2N/3$.

Implementation status & work completed

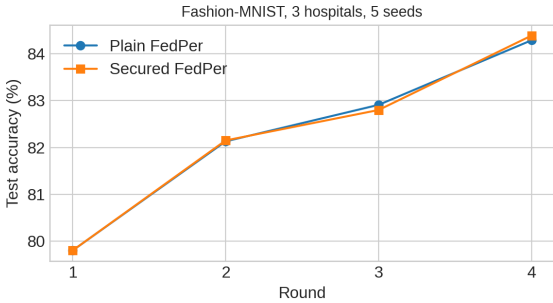
- **Core FL infrastructure:** FastAPI backend and React/Vite operations console; Flower with a custom MPC strategy (FedAvg / FedPer).
- **Post-quantum security:** ML-DSA-65 enrollment and per-round signatures, ML-KEM-768 pairwise agreement, Shamir custody (pure-Python reference implementations).
- **Finite-field MPC:** GF($2^{40} - 87$) masking, self-masks, hash ratchet and Shamir dropout recovery.
- **TCR–epitope modelling:** VDJdb six-stage pipeline, clonotype-disjoint split, Top-10 classifier and Siamese FastKAN, now wired into the FL loop (previously pending).
- **Validation:** Fashion-MNIST utility check, MITM / forward-secrecy / naive-masking tests, membership-inference benchmark.
- **Reproducibility:** 10-seed claims ledger (commit `7d9d12e`), tag `v1.3-audit-final`.

Experimental setup and data pipeline

Stage	Records	Retained
Raw VDJdb (all species, chains)	284,546	100%
Human host	261,261	91.82%
TRB chain	149,366	52.49%
Confidence score ≥ 1	15,814	5.56%
Top-10 abundant epitopes	10,260	64.88%
Ambiguous CDR3s removed	9,420	91.81%
Unique clonotypes (CDR3)	3,864	80/20 split

- 3 hospitals, Dirichlet $\alpha = 0.5$.
- 10 seeds (42–51), 10 rounds, fixed budget for all models.
- 3,091 train / 773 test clonotypes, no overlap.
- Utility check: Fashion-MNIST, 5 seeds.

Results: utility preservation of secured aggregation



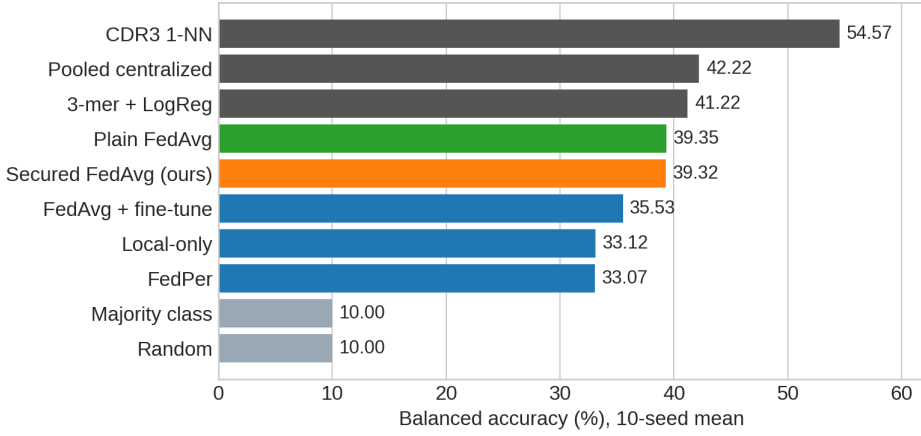
- Final-round accuracy: plain 84.29%, secured 84.39%.
- Delta 0.0953 ± 0.0634 pp (paired 95% CI $[+0.01, +0.18]$ pp).
- One-round equivalence test: $\max|\Delta\theta| \approx 2.4 \times 10^{-7}$, below the 10^{-6} quantization step.

Results: Top-10 benchmark (10 seeds)

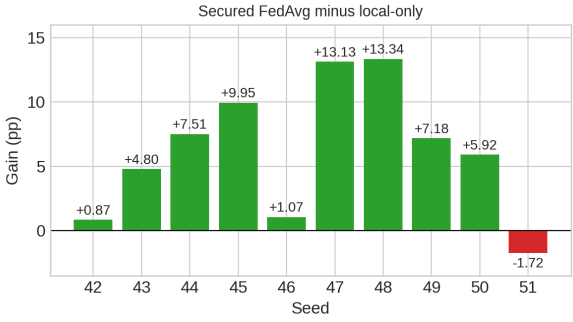
Method	Bal. acc (%)	Raw acc (%)	Macro F1	Macro AUROC
Random / majority class	10.00	10.00 / 46.04	0.100 / 0.063	0.500
CDR3 1-NN (Levenshtein)	54.57±2.65	61.23±3.34	0.500	–
3-mer + logistic regression	41.22±4.68	57.76±4.87	0.431	0.829
Pooled centralized reference	42.22±4.96	55.63±5.10	0.418	0.832
Local-only (mean of 3 hospitals)	33.12±3.33	46.88±6.59	0.291	0.754
Plain FedAvg + FastKAN	39.35±5.79	56.40±5.56	0.384	0.817
Secured FedAvg + FastKAN (ours)	39.32±5.75	55.90±5.58	0.373	0.819
FedAvg + local fine-tuning	35.53±5.08	49.31±7.52	0.329	0.806
FedPer (global test)	33.07±6.86	49.80±10.44	0.311	0.802

Secured and plain FedAvg are statistically indistinguishable, so masking costs no utility.

Results: balanced accuracy by method

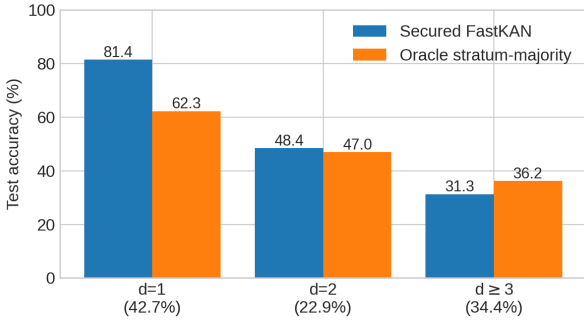


Results: federation beats isolated hospitals



- Mean paired gain $+6.21 \pm 5.11$ pp (39.32% vs 33.12%).
- 95% CI $[+2.55, +9.86]$ pp; paired $t = 3.84$, $p = 0.00397$.
- Positive in 9 of 10 seeds; seed 51 is the exception (-1.72 pp).
- Isolated hospitals collapse to the regional dominant epitope: 46.88% raw but 33.12% balanced accuracy.

Results: generalization by sequence distance

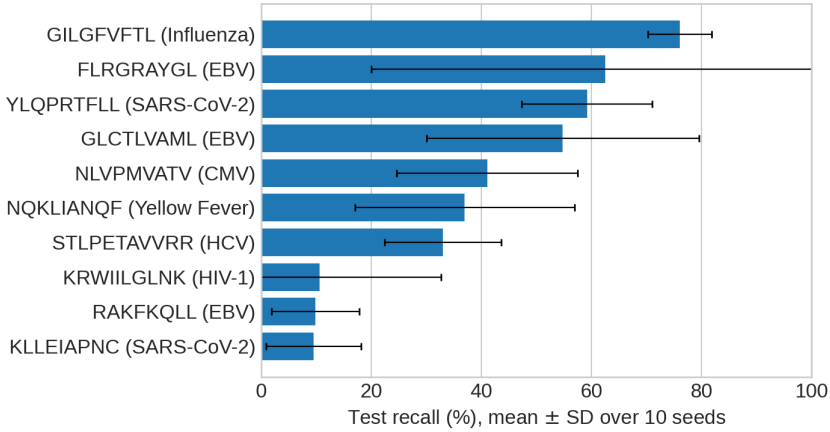


Stratum	Δ vs oracle	p
$d = 1$	$+19.15$ pp	0.0016
$d = 2$	$+1.44$ pp	0.18
$d \geq 3$	-4.93 pp	0.0047

- 65.6% of test TCRs have $d \leq 2$.
- Skill sits in near-homologs.

The oracle takes each stratum's modal label from the test labels; the non-oracle global majority reaches 46.04%.

Results: per-epitope recall (plain FedAvg)



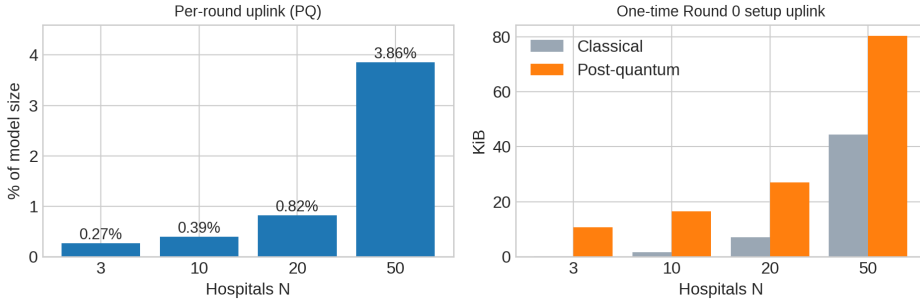
Results: post-quantum primitive benchmark

Metric	Classical	Post-quantum	When paid
Key-agreement public key	32 B (X25519)	1,184 B (ML-KEM-768)	Once, Round 0
Ciphertext	32 B	1,088 B	Once, Round 0
Signature	64 B (Ed25519)	3,293 B (ML-DSA-65)	Per update
Verification key	32 B	1,952 B	Once, Round 0
Signing latency*	0.031 ms	51.83 ms	Per update
Verification latency*	0.089 ms	11.22 ms	Per update
Quantum resistance	None (Shor)	NIST Category 3	–

*Classical uses compiled OpenSSL; post-quantum uses pure-Python reference code (100 trials). Optimized C (liboqs) signs in under a millisecond. FIPS 204 signature: 3,309 B.

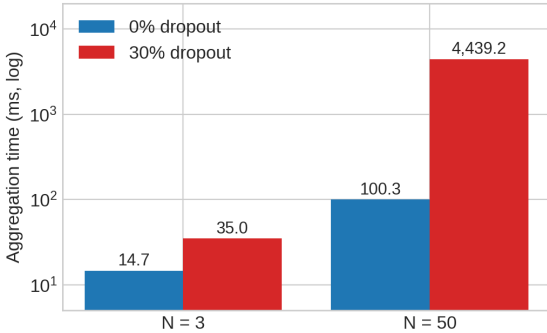
Steady-state post-quantum increment over classical: +3.15 KiB per round (the signature delta).

Results: communication overhead



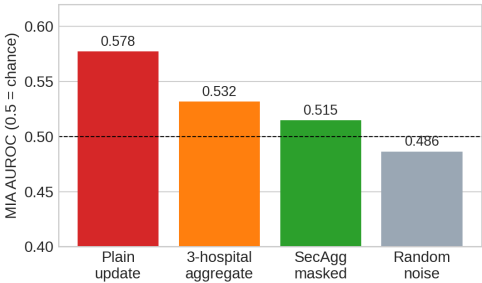
- At $N = 50$: 47.48 KiB per round (3.86% of the 1.26 MB model); 50.59 KiB (4.11%) with self-mask shares and AEAD framing.
- One-time Round 0 downlink at $N = 50$: 214.23 KiB (17.40%). Shares grow as $O(N^2)$ cohort-wide.

Results: dropout recovery cost



- 0% dropout: 14.7 ms ($N = 3$), 100.3 ms ($N = 50$).
- 30% dropout: 35.0 ms ($N = 3$), 4,439.2 ms ($N = 50$): a 44.3× slowdown.
- The server rebuilds $k(N - k)$ stranded pairwise seeds by interpolation and re-expands their masks.

Results: security and attack validation



- Membership inference, 20 seeds: plain update AUROC 0.578 ± 0.048; masked 0.515 ± 0.061 (CI spans 0.5).
- Paired reduction $+0.0625$, $p = 0.0012$; masked update is indistinguishable from random noise.
- Revealed 3-hospital aggregate: 0.532, so it needs differential privacy.
- MITM ciphertext substitution: signature check fails (pass).
- Forward secrecy: round-2 seed cannot unmask round 1 (pass).
- Public-seed masking is recoverable; ML-KEM masking is not (pass).

Earlier results: synthetic and open-world retrieval

Experiment	Result	Reading
Synthetic motifs, 15 epochs	Top-1 97.25%, AUROC 0.995	Validates gradient flow only
VDJdb 485-way, 50 epochs	Top-1 2.8% (chance 0.2%), AUROC 0.850	Above chance, low in absolute terms
Open-world, unseen-TCR split	$\approx$0% balanced Top-1; majority collapse (37.26% raw)	450+ epitopes have fewer than 10 samples

485-way ablation (30 epochs)	Loss	Top-1 (%)	AUROC	MRR
FedPer + FastKAN	3.69	1.04	0.815	0.049
FedPer + MLP	3.78	0.68	0.813	0.047
FedAvg + FastKAN	4.25	0.68	0.777	0.034

Ablation values are descriptive (no uncertainty estimates). Closed Top-10 classification is the right regime when training from scratch; open-world retrieval needs pretrained embeddings.

Discussion and significance

- **Security:** authenticated ML-KEM-768 + ML-DSA-65 closes MITM and quantum-eavesdropping gaps; self-masks and ratcheted seeds make dropout recovery forward-secret, for +3.15 KiB per round over classical signatures.
- **Utility:** aggregation equivalence $\approx 2.4 \times 10^{-7}$; Fashion-MNIST delta 0.095 pp; +6.21 pp over isolated hospitals ($p = 0.004$).
- **Honest baselines:** CDR3 1-NN (54.57%) and pooled training (42.22%) still beat secured FedAvg (39.32%) on balanced accuracy; skill is concentrated in near-homologs.
- **FedAvg over FedPer:** global averaging regularizes rare-epitope learning under Dirichlet skew.
- **Scope:** simulated three-hospital setting on public VDJdb; not clinical evidence. Pure-Python crypto timings are upper bounds.

Conclusion

- **Integrated design:** ML-DSA-65 authentication, ML-KEM-768 key agreement, self-masked GF($2^{40} - 87$) aggregation and forward-secret dropout recovery, implemented and benchmarked end to end.
- **Two-stage application:** Fashion-MNIST validates the pipeline; VDJdb Top-10 classification evaluates TCR–epitope learning.
- **Measured benefit:** utility preserved under masking; federation gains 6.21 pp over local training.

Next milestones

- Differential privacy on the revealed aggregate; norm clipping and range proofs against poisoning.
- Pretrained protein-language-model embeddings (ESM-2, ProtBERT) for distant and rare epitopes.
- Audited C implementations (liboqs), real multi-node deployment, signed survivor-consistency rounds.

References I

[1] H. B. McMahan et al., "Communication-efficient learning of deep networks from decentralized data," AISTATS, 2017.

[2] M. G. Arivazhagan et al., "Federated learning with personalization layers," arXiv:1912.00818, 2019.

[3] K. Bonawitz et al., "Practical secure aggregation for privacy-preserving machine learning," ACM CCS, 2017.

[4] NIST, FIPS 204: Module-Lattice-Based Digital Signature Standard, Aug. 2024.

[5] NIST, FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard, Aug. 2024.

[6] X. Zhang et al., "PQSF: Post-quantum secure privacy-preserving federated learning," Sci. Rep. 14:23553, 2024.

References II

[7] Y. Li et al., "Secure and efficient multi-key aggregation for federated learning," Inf. Sci. 654, 2024.

[8] L. Chen et al., "Secure and efficient federated learning via novel multi-party computation and compressed sensing," Inf. Sci. 667, 2024.

[9] P. Ren et al., "CosPer: An adaptive personalized approach for enhancing fairness and robustness of federated learning," Inf. Sci. 675, 2024.

[10] G. Croce et al., "Deep learning predictions of TCR-epitope interactions reveal epitope-specific chains in dual alpha T cells," Nat. Commun. 15:3211, 2024.

[11] L. Su et al., "DKD-pFed: personalized federated learning via decoupling knowledge distillation and feature decorrelation," Expert Syst. Appl. 259, 2025.

[12] D. Commey and G. V. Crosby, "PQS-BFL: A post-quantum secure blockchain-based federated learning framework," Expert Syst. Appl. 312, 2026.

References III

[13] P. Rajabi and M. Toorani, "Secure aggregation for privacy-preserving federated learning on clinical EEG data," arXiv:2607.28191, 2026.

[14] P. Nayak et al., "Post-quantum federated learning: secure and scalable threat intelligence for collaborative cyber defense," arXiv:2603.07726, 2026.

[15] Y. Zhang et al., "Beskar: Efficient full-stack private federated deep learning with post-quantum security," IEEE TDSC, arXiv:2505.05751, 2025.

[16] H. Zhang et al., "Enhancing quantum security over federated learning via post-quantum cryptography," IEEE TPS-ISA, arXiv:2409.04637, 2024.

[17] J. Bell et al., "Secure single-server aggregation with sub-linear communication," ACM CCS, 2020.

[18] D. J. Beutel et al., "Flower: A friendly federated learning research framework," arXiv:2007.14390, 2020.

References IV

[19] M. Shugay et al., "VDJdb," Nucleic Acids Res. 46(D1), 2018; M. Bagaev et al., "VDJdb in 2021," 49(D1), 2021.

[20] A. Shamir, "How to share a secret," Commun. ACM 22(11), 1979.

[21] Z. Li, "Kolmogorov–Arnold networks are radial basis function networks," arXiv:2405.06721, 2024.

[22] A. van den Oord et al., "Representation learning with contrastive predictive coding," arXiv:1807.03748, 2018.

[23] L. Zhu et al., "Deep leakage from gradients," NeurIPS, 2019.

[24] H. Xiao et al., "Fashion-MNIST," arXiv:1708.07747, 2017.

[25] United Nations General Assembly, "Transforming our world: The 2030 Agenda," A/RES/70/1, 2015.